

Superset Determinations in Modal Triplet Theory: Parameter Identifiability after True-SM Closure

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Abstract

We reformulate the MTT superset program as an audited parameter-identifiability framework. The former one-loop gauge crossings, fitted ζ ratios, minimum-norm threshold vector, common scale K , and claimed $\alpha_s(M_Z)$ cross-prediction are withdrawn. Their inputs and scale choices were not independent. The current authority instead closes embedded renormalized-Standard-Model equivalence at a one-shared-physical-primitive, measured-profile standard, using selected SMDR v1.3 transport and a complete eight-row covariance. We classify every output as structural theorem, calibration, replay/profile coordinate, diagnostic, or held-out prediction. This preserves the superset strategy as a rigorous tool for comparing encodings while preventing fitted values from being promoted into source theorems.

Revision note for this edition

Supersedes. *Superset Determinations in Modal Triplet Theory*, version 2.

Reason. One-loop crossings, fitted zeta ratios and threshold vectors, and the claimed $\alpha_s(M_Z)$ cross-prediction reused target-related data and were not independent determinations.

Resolution. Version 3 replaces the numerical claim chain by a parameter- identifiability ledger tied to the selected SMDR profile and classifies every quantity by theorem, calibration, replay, diagnostic, or prediction status.

Retained result. The superset strategy remains useful for comparing encodings and discovering shared-source constraints.

Remaining boundary. Predictive promotion requires independently selected source parameters and genuinely held-out observables.

1 Purpose of the superset program

MTT can organize several effective encodings through shared carrier, projector, bundle, overlap, and response data. The superset strategy is useful when it asks which observables are jointly representable and which parameters are identifiable from a declared input set. It is not, by itself, a method for deriving measured constants.

We use five disjoint status classes:

1. **Structural theorem:** follows from selected mathematical data without observed value selection.
2. **Calibration:** parameters are chosen to reproduce declared data.
3. **Profile/replay:** measured coordinates are transported through a specified scheme or embedded action.

4. **Diagnostic:** a comparison tests consistency but was involved in selection or fitting.
5. **Held-out prediction:** the compared observable was absent from source selection, calibration, branch choice, and uncertainty construction.

2 Retirement of the legacy Tier-3 execution

The former scales Λ_{12} and Λ_{23} were obtained from a one-loop crossing exercise. The reported values near 4.2–5 TeV were then used to extract ζ ratios, fit a common scale, choose a minimum-norm threshold, and reconstruct α_s . This chain is retired.

Specifically:

- a gauge crossing is not an MTT coherence or cutoff scale;
- the old 0.560 and 0.229 ratios are not current profile values;
- K was calibrated with gravitational/electroweak information and was not an independent prediction;
- the minimum-norm threshold was a regularized fit selected by an external norm, not a source theorem; and
- α_s was not held out from the latent extraction and threshold choices, so it was not a cross-prediction.

3 Current common-scheme profile

The selected numerical transport uses SMDR v1.3 at

$$Q = M_t = 172.5590883453979 \text{ GeV}.$$

Its gauge entries are

$$g_Y = 0.3585945042 \pm 0.0000307251, \quad g_2 = 0.6475986708 \pm 0.0000287665, \quad g_3 = 1.163427409 \pm 0.004036156.$$

Fifteen locked source coordinates are transported into eight full-SM $\overline{\text{MS}}$ rows. The differentiated map emits a positive-definite 8×8 covariance with all 36 symmetric entries and all 15 BCT–WZH cross-block entries. These rows are profile transport, not MTT-derived constants.

4 Adopted closure and parameter count

The final global audit closes 12/12 obligations for embedded renormalized-SM equivalence at the adopted standard. The accounting is:

- one shared physical primitive P_{EW} , counted once;
- zero Higgs-specific adjustable parameters at the embedding layer;
- measured threshold, mass-scheme, and magnitude/profile coordinates retained explicitly as profile data; and
- imported standard SM perturbative quantization and observable transport.

This is a more economical cross-sector representation than independently retuning each sector, but it is not zero-input physics. Profile coordinates must not be hidden when comparing parameter counts with the Standard Model.

The magnitude-bearing profile contains ten scalar labels: nine charged Yukawa magnitudes and λ_H . At the strict internal source tier, accepted values remain 0/10. Seven physical threshold

and three mass-scheme rows are likewise accepted at the admitted profile tier while strict internal emitted values remain zero.

5 Selected structural outputs

The current superset contains genuine source-side structure independent of the profile values, including:

- the locked 27-by-27 qutrit–Weyl/minimal matrix architecture;
- the selected rank-two literal Cech–HYM witness, closed 2/2;
- the internal $K_{\text{threshold}}$ response ledger, closed 10/10;
- the selected retarded $q79/F/m1$ orientation representative; and
- the CKM II rows, closed at the prediction-with-uncertainty-profile standard.

These achievements constrain how the measured profile can be embedded. They do not automatically emit every profile magnitude.

6 Strict-upgrade ledger

Beyond the adopted baseline, nine stronger obligations are tracked:

ID	Upgrade	Status
U1	zero-primitive empirical source	open
U2	literal global Cech–HYM/QaSU3	closed 2/2
U3	official joint input likelihood	partial
U4	CKM prediction beyond source rows	closed at profile criterion
U5	absolute neutrino mass and ontology	partial
U6	strong-CP selection	partial
U7	MTT-derived quantization	partial
U8	constructive nonperturbative 4D QFT	partial
U9	unique observed-branch selection	partial

Closing U2 or U4 does not imply that the other seven upgrades close.

7 Identifiability theorem

Theorem 7.1 (Profile-standard identifiability). *Given the selected branch, common-scheme profile, one shared electroweak primitive, embedded SM action, and selected finite operator data, the accepted observable rows are jointly reproducible with their declared covariance. Removing the measured magnitude/threshold profile destroys numerical identifiability of those rows because the current internal source inventory emits no replacement scalar values.*

Proof. The selected transport and embedding provide the forward map and covariance at the profile standard. The strict source-row audits attempt all required scalar slots and accept zero internal magnitude and physical threshold/mass values. Therefore the profile rows are sufficient for the adopted forward execution and presently necessary for its numerical values. $\square$

8 Rules for future superset calculations

Every future execution must publish raw inputs, units, scheme and scale, selection rules, fitted parameters, covariance, branch choices, code, and unit tests. A claimed held-out prediction must include a data-separation certificate showing that the observable was absent from all source and calibration paths. No matching point, internal spectral gap, or gauge crossing may be promoted to a physical cutoff without a separate source theorem.

9 Conclusion

The superset program survives, but in a sharper form. Its current success is the coherent, reproducible embedding of the renormalized SM and the reduction of many sector-specific constructions to shared finite operator data. The old crossing-scale calibration and its downstream numerical claims are retired.

The next frontier is not another backfit. It is source promotion: emit the ten magnitude rows and physical threshold/mass values from selected MTT data, or retain them transparently as measured profile coordinates. That distinction is the correct measure of progress toward strict no-knob closure.

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